Consumer Confidence and the U.S. Economy

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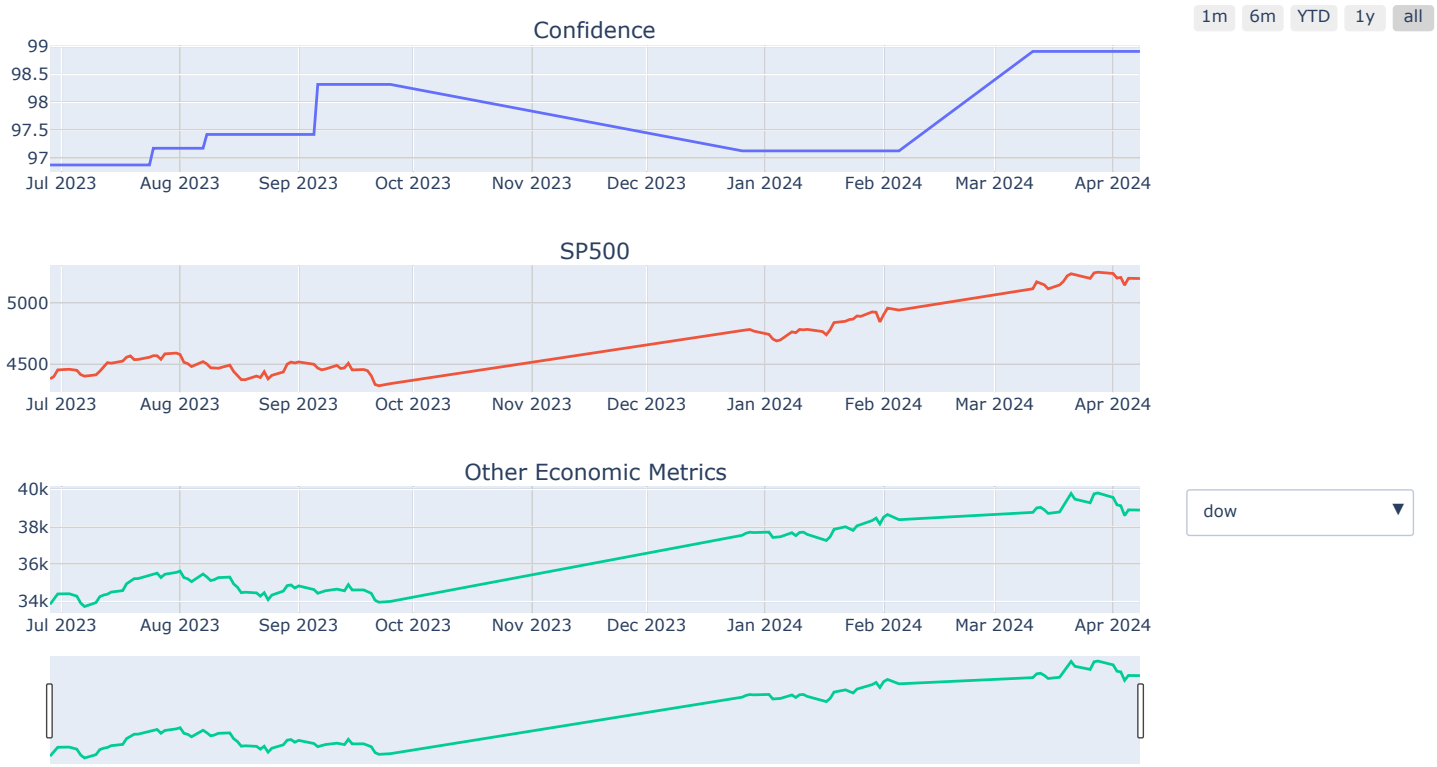


Metric	Description
Confidence	Consumer Confidence
SP500	S&P 500
dow	Dow Jones Industrial Average
nasdaq	NASDAQ Composite
inflation_5yr	5-Year Breakeven Inflation Rate
oil	Crude Oil Prices: West Texas Intermediate
interest	Federal Funds Effective Rate
t_note_10yr	US Treasury Securities - 10 year
cpi_item_cost	Consumer Price Index - avg price
cpi_purchase_power	Consumer Price Index - purchase power

Overview:
The project's goal is to visually compare consumer confidence of the US economy with respect to important economic metrics from the Federal Reserve's FRED API using Dash and Plotly. Furthermore, the project showcases my knowledge of data, deployment and cloud tools.

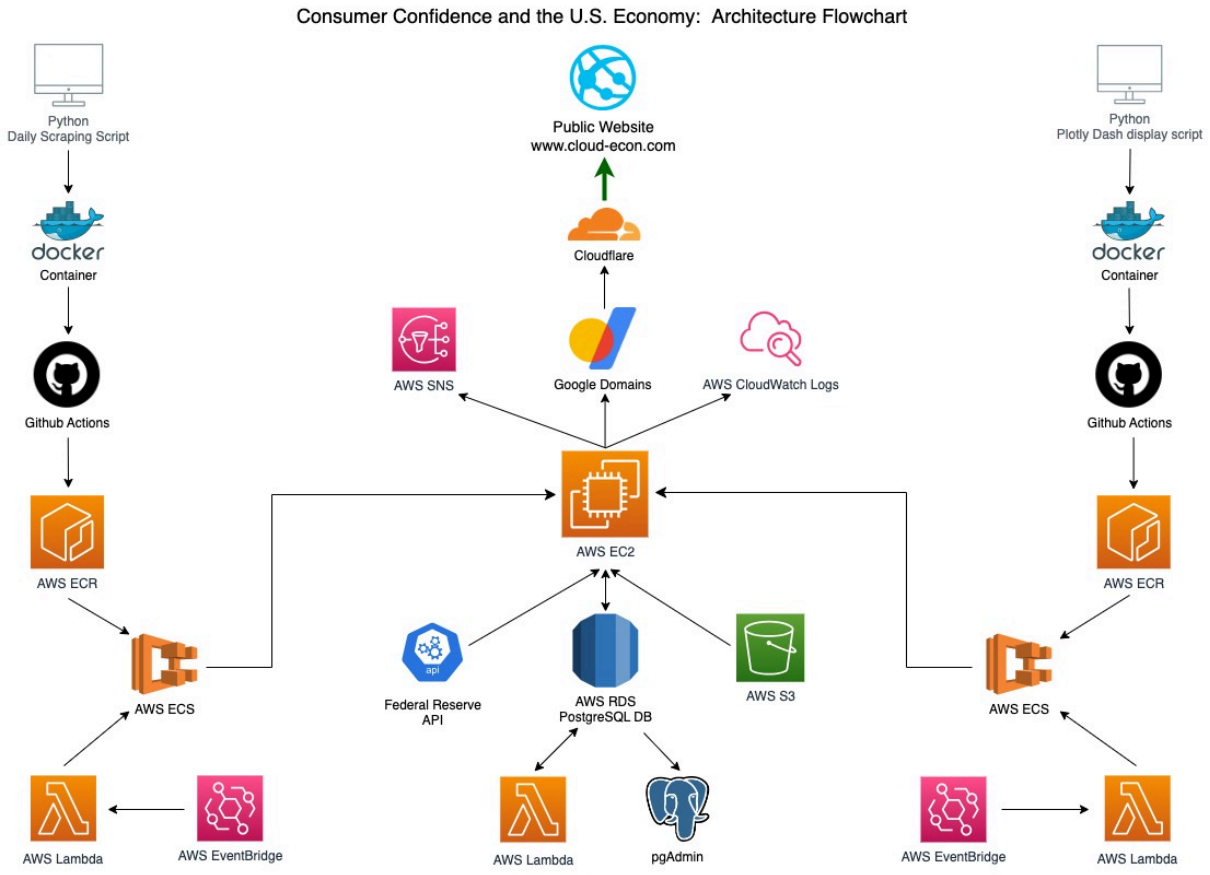
Initial questions to explore:
What is the current consumer confidence of the economy?
How has it varied over time?
Is there a correlation between consumer confidence and economic metrics like stock market performance?

Once enough data are collected, they will be fed into machine learning models to best fit a predictive model and determine if any economic indicators have correlations with consumer confidence as the target variable.

Methodology:
1. Download economic data from U.S. Federal Reserve's FRED API and create a pandas dataframe.
2. Run Dockerized python script daily to update a AWS RDS PostgreSQL database using AWS ECS automation.
3. Visualize data with Plotly Dash and deploy as a website using AWS ECR, ECS, EC2 & Lambda.

Skills:
Python; FRED API; Pandas; Docker; Github Actions; PostgreSQL; AWS - RDS, S3, ECR, ECS, EC2, Lambda, SNS, Cloudwatch, IAM, EventBridge; Plotly; Dash; pgAdmin; YAML; Cloudflare; Data Munging; Automation; Continuous Delivery

Architecture Flowchart:



Resources:
Github portfolio
https://github.com/lflrnr/Portfolio/
Federal Reserve Bank of St. Louis - FRED API docs
https://fred.stlouisfed.org/docs/api/fred/

Disclaimers:
"This product uses the FRED® API but is not endorsed or certified by the Federal Reserve Bank of St. Louis." Furthermore, this exercise is for demonstration purposes only. This dashboard, the underlying code, technologies involved and their related organizations and the author does not assume any liability for the use or misuse of any of the aforementioned resources nor claim their veracity, timeliness, or conclusions stated or inferred.

Confidence metric citation:
Organization for Economic Co-operation and Development, Consumer Opinion Surveys: Confidence Indicators: Composite Indicators: OECD Indicator for the United States CSCICP03USM665S , retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/series/CSCICP03USM665S, August 4, 2023.

Endnotes:
Initially, this project was to conduct sentiment analyses of the US economy using Twitter data. Coding was completed but Twitter drastically reduced its free tier rate limits that were needed for the code to work. An example run is posted at my Github portfolio found above. Finally, the latest available date displayed my not be recent due to FRED having lags in data updates which usually resolve within a week or two.